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hw2_me501_Q4.m

(C) Andrew Sabelhaus, 2026

```
clear all;
close all;
clc;

disp('Practice with eigenvector and eigenvalue calculations.')

Practice with eigenvector and eigenvalue calculations.
```

Setup: our three matrices

```
B = [1 1 0;
     0 2 -1;
     0 1 2];

C = [2 1 0;
     1 2 0;
     0 0 3];

G = [2 1 0;
     -1 4 0;
     0 0 1];
```

Calculation of eigenvectors and eigenvalues:

```
[VB, DB] = eig(B)
[VC, DC] = eig(C)
[VG, DG] = eig(G)
```

VB =

```
1.0000 + 0.0000i    0.3162 - 0.3162i    0.3162 + 0.3162i
0.0000 + 0.0000i    0.6325 + 0.0000i    0.6325 + 0.0000i
0.0000 + 0.0000i    0.0000 - 0.6325i    0.0000 + 0.6325i
```

DB =

```
1.0000 + 0.0000i    0.0000 + 0.0000i    0.0000 + 0.0000i
0.0000 + 0.0000i    2.0000 + 1.0000i    0.0000 + 0.0000i
```

$$0.0000 + 0.0000i \quad 0.0000 + 0.0000i \quad 2.0000 - 1.0000i$$

$VC =$

$$\begin{array}{ccc} -0.7071 & 0.7071 & 0 \\ 0.7071 & 0.7071 & 0 \\ 0 & 0 & 1.0000 \end{array}$$

$DC =$

$$\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{array}$$

$VG =$

$$\begin{array}{ccc} 0.7071 & -0.7071 & 0 \\ 0.7071 & -0.7071 & 0 \\ 0 & 0 & 1.0000 \end{array}$$

$DG =$

$$\begin{array}{ccc} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{array}$$

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